Technical Report

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Results

Our Final First Stage Model is as follows:

$P(AtRisk) = \beta_0 + \beta_1 YearsofEducation + \beta_2 ADD/ADHD + \beta_3 Gender : Male + \beta_4 Offense : Other + \beta_5 Offense : Property + \beta_6 Offense : PublicOrder + \beta_7 Offense : Unknown + \beta_8 Offense : Violent + \beta_9 Children + \beta_{10} TimesArrested + \beta_{11} MentalHealthTreatment + \beta_{12} MentalHealthTreatment$

FIXXXXXX LATER After conducting a Likelihood-Ratio Test on our model, we find that our final model has a deviance score of 4685 on 12 degrees of freedom and a p-value less than 0.001. This model was fit on $n = 15,450$ observations with 1708 dropped/deleted observations. Over 90% are due to alcoholism.

This indicates our full model significantly outperformed the null model and the combination of clinical and demographic factors provides substantial predictive power for inmate infraction risk. Overall, our model had a 73% classification accuracy (classification of inmates who had infractions or not)

Table 1: Stage 1 Confusion Matrix

Classification	Observed: No Infraction (0)	Observed: At Risk (1)
Predicted: Low Risk (0)	5,361	2,365
Predicted: High Risk (1)	1,755	5,969

Note:
Overall Accuracy: 73.3% | Sensitivity: 75.3% | Specificity: 71.6% | P-Value: $<$ .001 | No Information Rate (NIR): 53.9%

The model diagnostics state that the most significant predictors are ADD/ADHD, Male Gender, Times Arrested, and Mental Health Treatment. Out of these, the predictors with the largest effect size are ADD/ADHD with 1.33, Male Gender with -0.938, and Mental Health Treatment with 2.201.

Table 2: Significant Predictors

Variable	Estimate	OR	Z_Value	95% Confint OR
ADD/ADHD	1.3389959	3.8152109	28.22076	(3.48 - 4.19)
Male Gender	-0.9386538	0.3911541	-19.65422	(0.36 - 0.43)
Mental Health Treatment	2.2016663	9.0400643	43.74620	(8.2 - 9.98)

The final Second Stage Model is as follows:

$ln(Infractions) = \beta_0 + \beta_1 Race : Black + \beta_2 Race : Hispanic + \beta_3 Race : Multiracial + \beta_4 Race : Other + \beta_5 Race : White + \beta_6 Number of Diagnoses + \beta_7 Armed Forces + \beta_8 Times Arrested + \beta_9 Years of Education + \beta_{10} ADD/ADHD + \beta_{11} Homeless + \beta_{12} Children + \beta_{12} Mental Health Treatment + \beta_{13} Alcoholism + \beta_{14} Alcohol Treatment + \beta_{15} Gender : Male + \beta_{16} + \beta_{17} Offense : Other + \beta_{18} Offense : Property + \beta_{19} Offense : Public Order + \beta_{20} Offense : Unknown + \beta_{21} Offense : Violent + \beta_{22} Number of Diagnoses * Mental Health Treatment + \beta_{23} Alcoholism * Alcohol Treatment$

Comparing the 2 models using their goodness-of-fit statistics, it is seen that a negative binomial model is more favorable with a lower AIC and Overdispersion statistic (ϕ). The negative binomial model was fitted on the subpopulation of $n = 7,619$ of those rated as "High Predicted Risk." This was determined by our previous logistic model of those who had a higher probability of having an infraction than the median value of the data set, which was 0.497. 105 observations were not counted, over 90% were deleted due to Race being unavailable.

Table 3: Model Comparison

Model	AIC	Deviance	df	Overdispersion
Poisson	35484.49	2373.456	23	8.3143945
Negative Binomial	19761.59	385.641	23	0.2250685

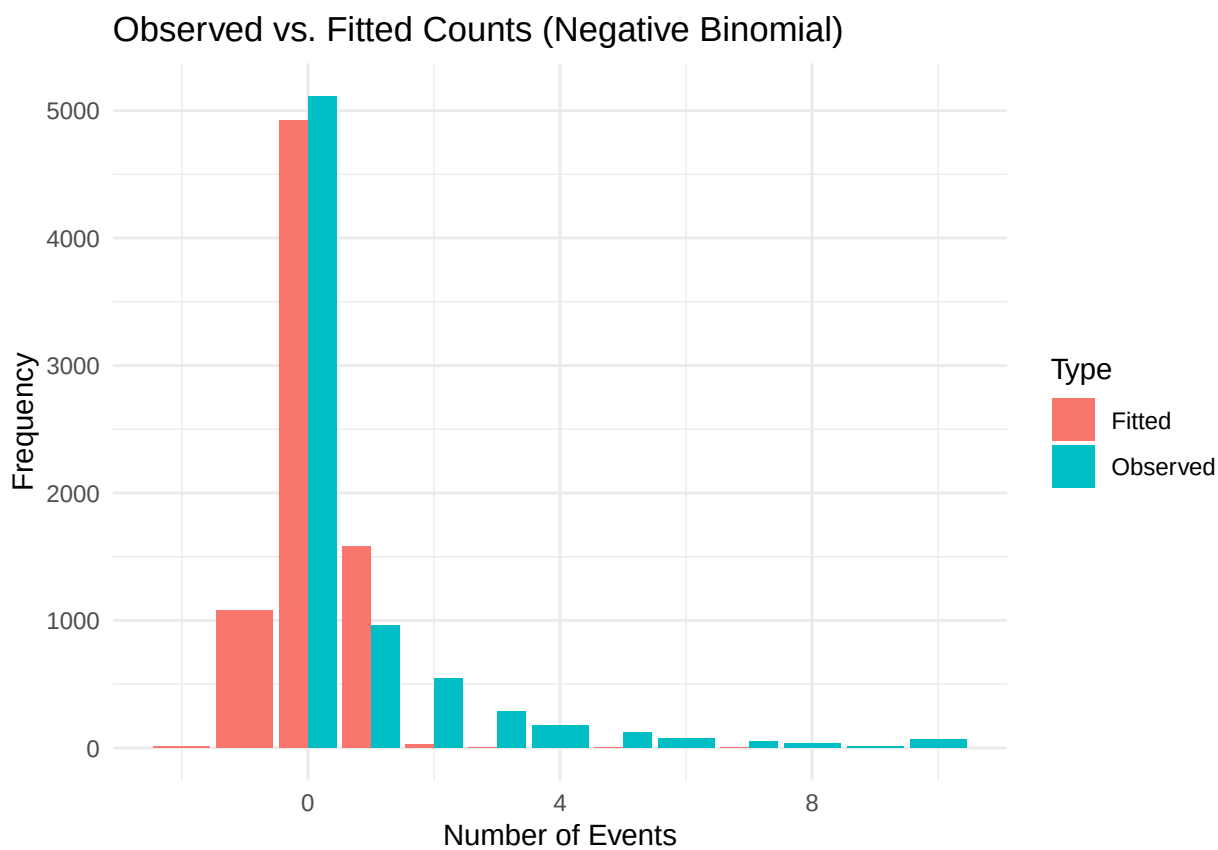
In our model, the most significant predictors are ADD/ADHD, Times Arrested, Children, Mental Health Treatment, Alcoholism, and an Unknown Offense Type. The variables with the largest effect size were ADD/ADHD with a 0.605 Coefficient and Unknown Offense type with -0.513 coefficient.

Table 4: Model Statistics

Variable	Estimate	Z.Value	P.Value
ADD/ADHD	0.6047363	9.484893	0.0000000
Times Arrested	0.0079125	6.885432	0.0000000
Children	-0.3953537	-6.714455	0.0000000
Mental Health Treatment	0.3248295	3.468970	0.0005225
Alcoholism	0.2941803	3.608199	0.0003083
Offense Type: Unknown	-0.5126795	-3.840658	0.0001227

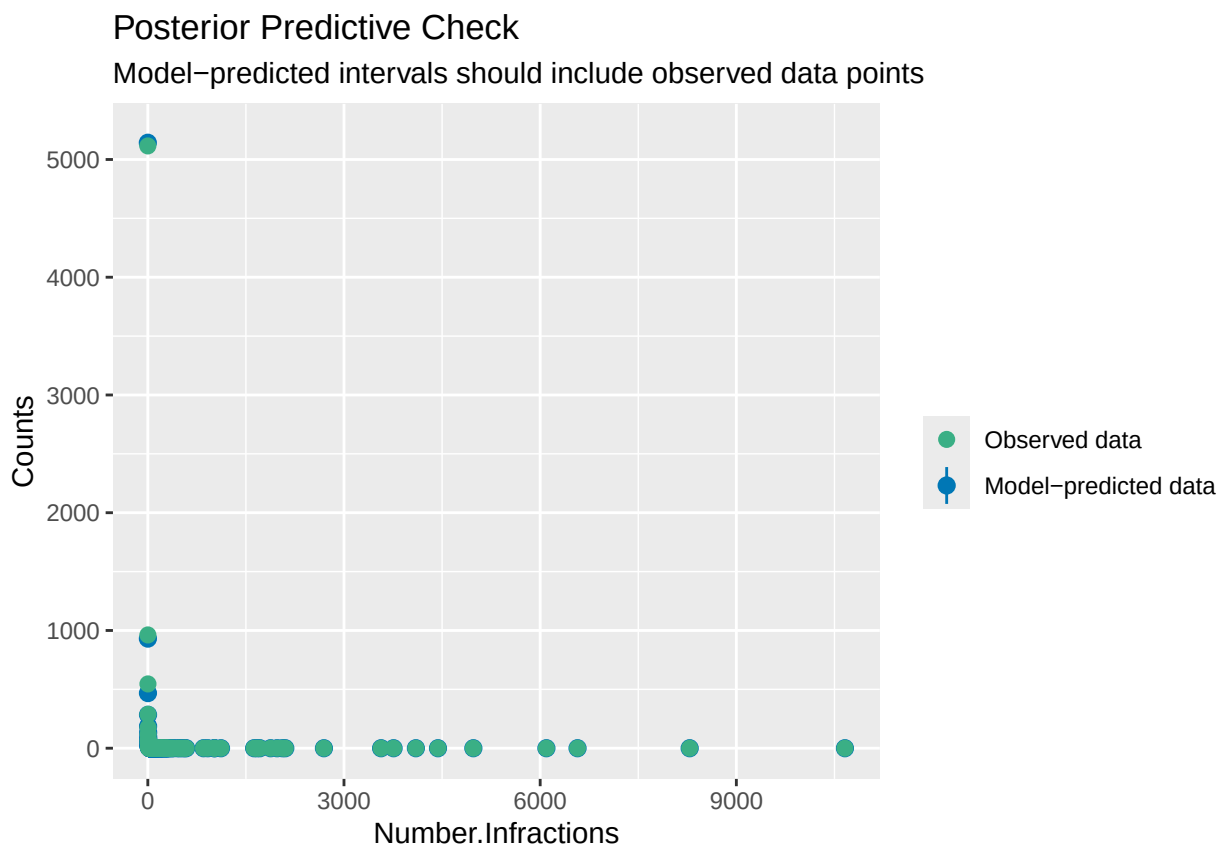
```
fittedN<-model.frame(stage2_model)


count_comp <- data.frame(
  Observed = fittedN$Number.Infractions,
  Fitted = predict(stage2_model)
) %>%
  # Round fitted values to nearest integer to compare counts
  mutate(Fitted = round(Fitted)) %>%
  gather(Type, Value) %>%
  filter(Value <= 10) # Filter to low counts (where most action happens)

# 2. Plot the distributions
ggplot(count_comp, aes(x = Value, fill = Type)) +
  geom_bar(position = "dodge") +
  labs(
```

```r
    title = "Observed vs. Fitted Counts (Negative Binomial)",
    x = "Number of Events",
    y = "Frequency"
  ) +
  theme_minimal()
```




```r
check_predictions(stage2_model)
```

```
## Ignoring unknown labels:
## * size : ""
## * alpha : ""
```


Posterior Predictive Check
Model−predicted intervals should include observed data points
Counts
5000
4000
3000
2000
1000
0
0
3000
6000
9000
Number.Infractions
Observed data
Model−predicted data